Balanced forces can go with steady motion

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Inertia / ES: inercia / PT: inércia

More mass resists velocity change more.

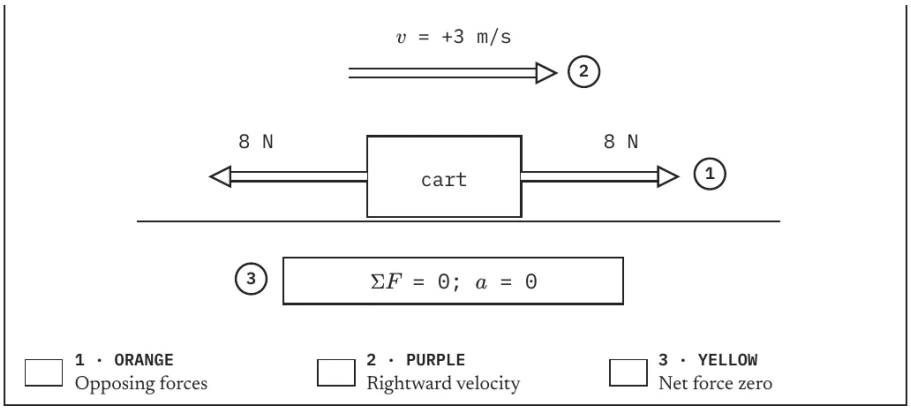
Also on your note page: Inertial frame, Net force, Equilibrium.

B READ THE RELATIONSHIP

Balanced forces → zero acceleration → constant velocity

As you read, find: balanced · steady · changes.

C TRACE THE ORIGINAL FIGURE



Trace equal forces first. Then trace velocity. In pencil, label the arrows F or v.

D FOLLOW THE WORKED EXAMPLE

A cart is at −2 m , moving at +3 m/s . Its external forces balance for the next 4 s . Find its final position.

Given x 0 = −2 m v i = +3 m/s Σ F = 0; t = 4 s

Rule First law: constant v . x f = x 0 + vt .

Work a = 0 m/s² x f = −2 + 3(4)

Answer +10 m Net force zero; motion continues.

E SAVE FOR THE FINAL MEETING

XJ is moving. A forward push is not needed to keep it moving. Gravity can still change its velocity.

Read the physics: Inertia

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Push a cart across a rough floor, then let go. It slows because friction still acts on it. The floor and the cart interact. Inertia is a property of the cart; it does not push.

02 HOW IT WORKS

A force is a push or pull. Net force means all the forces on one object added with their directions. When the forces balance, acceleration is zero. A moving cart keeps its speed and direction. A resting cart stays at rest.

03 WHAT TO CHECK

Use a frame that is not speeding up or turning. A purple velocity arrow shows motion. Orange arrows show forces. Never add the purple arrow to the orange arrows.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart is at −2 m , moving at +3 m/s . Its external forces balance for the next 4 s . Find its final position. +10 m Net force zero; motion continues.

Model check: Balanced forces → zero acceleration → constant velocity.

05 CONNECT TO XJ

XJ needs no engine to keep moving. Over a short interval, use steady velocity as a model. A gravity pull can change the speed or direction. A new observation can test the model.

SAY IT BEFORE YOU WRITE

Is the cart moving, or is its velocity changing? How can you tell? The cart moves ___. Its velocity ___ because the forces ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Inertia / ES: inercia / PT: inércia

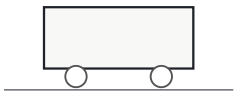
Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning in my words What does inertia mean? Say it the way you'd explain it to a friend. In my own words, inertia means …	**How can you tell?** An empty cart and a loaded cart both start at rest. We push each with the same net force for one second. Which gains less speed? Why? The ___ cart gains less speed because …
Example or sketch A cart is already rolling right. The forces on it balance. Where would you draw it one second later? And one second after that? Label your three carts 0 s, 1 s and 2 s.	**Non-example and why** Someone points to the hand pushing the cart and says, "That push is inertia." What would you tell them? The hand supplies ___. The cart has ___.

B DRAW WITH YOUR TEACHER

Draw a moving cart with balanced force arrows in orange and its separate velocity arrow in purple.

Labels you can use: balanced · steady · changes.

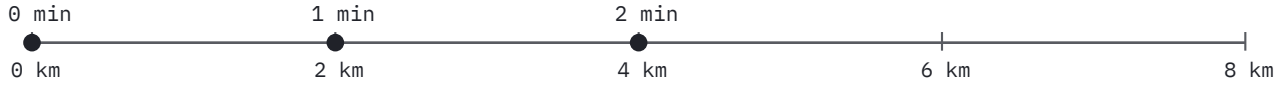


Tell a partner: The cart moves ___. Its velocity ___ because the forces ___.

Where should we point the telescope?

SECTION E / FINAL MEETING Name _____

The telescope spots XJ once a minute. At minute 3, the camera can take one picture. Your team gets to choose where it points.



Camera windows A +4 km B +6 km C +8 km

Choose a job together. One of you marks the prediction. The other makes the call.

☐ **Tracking engineer**

Circle the window you'd aim for. Tell the camera operator where to point and which two sightings helped you decide.

☐ **Live science broadcaster**

Tell viewers where to look. Someone asks, "If XJ has no engine, why does it keep moving?" Use your cart drawing to answer.

Try saying "Aim at ___ because ___." Or start your broadcast with "Watch window ___."

Mark your prediction here.



WHEN THE NEXT SIGHTING COMES IN

Mark the new sighting. How far did XJ move in the last minute? Switch speakers and tell your crew or your viewers what changed.

"We predicted ___. The telescope saw ___. In the last minute, XJ moved ___."

Why could a straight-line forecast work for a minute even while gravity pulls on XJ?